20260526 Notes: NE_SII and Te-based metallicity products in SFR+Z.py

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This note documents the current implementation in `SFR+Z.py` for the PyNeb electron-density product and the three Te-based oxygen-abundance products. The goal is to make the CANFAR products auditable: every finite value should be traceable to a specific detection gate, line ratio, PyNeb call, temperature assumption, and error-propagation step.

The three metallicity branches documented here are:

1. `O_H_BR24_DIRECT_{HII,SF}`: Brazzini et al. 2024 multi-zone direct-style method using measured Te([N II]) and measured Te([S III]).
2. `O_H_NII_OII7325_{HII,SF}`: MAUVE semi-direct ionic-sum method using measured Te([N II]), red [O II] 7325, and an inferred high-zone temperature.
3. `O_H_NII_K25_{HII,SF}`: Kreckel et al. 2025 / Mendez-Delgado et al. 2023 single-temperature Te([N II]) metallicity proxy.

All new abundance-like FITS maps use `BUNIT = 1`, not `12+log(O/H)`, so that CARTA/CASA treats them as dimensionless images rather than warning about an unknown FITS unit.

1. Region masks and detection gates

The script first applies the nGIST spatial mask when `v3tk_v7.6.8/{GALID}/{GALID}_mask.fits` exists. In the current MAUVE v3tk products, the script interprets:

```
MASK == 0  -> keep / usable spatial pixel
MASK == 1  -> masked spatial pixel
```

The mask is applied to `V_STARS2`, `SIGMA_STARS2`, and all raw gas-line flux and flux-error maps. Therefore later checks such as `np.isfinite(V_STARS2)` implicitly respect the spatial mask.

For Te products, an emission line is considered detected only if the raw gas-line maps satisfy:

$$\mathrm{detected}(F_\lambda) = \mathrm{finite}(F_\lambda) \land \mathrm{finite}(\sigma_\lambda) \land \sigma_\lambda > 0 \land F_\lambda / \sigma_\lambda \ge 3 \land F_\lambda \ge 20 .$$

The two numeric thresholds are the current script-level values:

```
cut   = 3
noise = 20
```

The Te products are computed twice:

- `_HII`: conservative Classification-2 mask, where SF means HII only.
- `_SF`: Classification-1 mask, where the existing script treats SF as HII+Composite in the [N II] BPT branch and HII in the [S II] BPT branch.

So if an HDU exists as `_HII`, the matching `_SF` HDU should also exist.

2. Dust correction used by all Te products

2.1 Balmer decrement

The observed Balmer decrement is:

$$BD = \frac{F_{\mathrm{H}\alpha}}{F_{\mathrm{H}\beta}} .$$

If `BD` is non-finite in a spatially usable pixel, the script sets it to the intrinsic value. If the corrected `BD` is below the intrinsic value, the script also floors it to the intrinsic value:

$$BD_{\mathrm{used}} = \max(BD, 2.86).$$

Thus the pipeline does not create negative gas reddening from `Halpha/Hbeta < 2.86`.

2.2 Gas E(B-V)

The local default extinction curve is Calzetti et al. 2000, matching the original MAUVE SFR+Z products. The code still contains an optional Milky Way CCM89 branch for explicit tests, but it is not the default. The current default is:

```
extinction_law = "calzetti"
mw_rv = 3.1  # used only if extinction_law is set to "mw"/"ccm89"
```

For each line, the script computes:

`k(\lambda) = \frac{A(\lambda)}{E(B-V)} .`

Then the Balmer-decrement reddening is:

`E(B-V)_{\mathrm{BD}} =`
`\frac{2.5}{k(\mathrm{H}\beta)-k(\mathrm{H}\alpha)}`
`\log_{10}`
`\left(`
`\frac{BD_{\mathrm{used}}}{2.86}`
`\right).`

This follows the standard Balmer-decrement form used in nebular work. Brazzini et al. 2024 and Kreckel et al. 2025 use an intrinsic Halpha/Hbeta ratio of 2.86 for their PHANGS-MUSE dust corrections, with an O'Donnell/CCM-family Milky Way curve. The local script deliberately keeps the original MAUVE convention and uses the Calzetti et al. 2000 curve by default for consistency with the previous SFR+Z products.

2.3 Corrected fluxes and errors

Each line flux is dereddened as:

`I_\lambda =`
`F_\lambda\,10^{0.4\,k(\lambda)\,E(B-V)_{\mathrm{BD}}}.`

The propagated uncertainty includes both the line-flux error and the Balmer-decrement reddening error:

`\sigma(I_\lambda)^2 =`
`\left[`
`\sigma(F_\lambda)\,10^{0.4\,k(\lambda)\,E(B-V)}`
`\right]^2`
`+`

$$
\left[
|I_\lambda|\,\ln(10)\,0.4\,|k(\lambda)|\,\sigma(E(B-V))
\right]^2 .
$$

The script now writes corrected flux and corrected error maps for every available gas line, with global, `_HII`, and `_SF` variants.

3. Electron density from [S II] 6716/6731

3.1 Required lines

The density map requires both [S II] lines to pass the detection gate:

```
SII6716_FLUX and SII6716_FLUX_ERR
SII6730_FLUX and SII6730_FLUX_ERR
```

The valid-density mask is:

$$
M_{n_e} =
\mathrm{detected}([\mathrm{S\,II}]6716)
\land
\mathrm{detected}([\mathrm{S\,II}]6731).
$$

3.2 Density ratio

The density-sensitive ratio is computed from dust-corrected fluxes:

$$
R_{\mathrm{SII}} =
\frac{I(6716)}{I(6731)} .
$$

The line-ratio uncertainty is the standard independent-error ratio formula:

$$
\sigma(R_{\mathrm{SII}}) =
|R_{\mathrm{SII}}|
\sqrt{
\left(\frac{\sigma_{6716}}{I_{6716}}\right)^2
+
\left(\frac{\sigma_{6731}}{I_{6731}}\right)^2
}.
$$

3.3 PyNeb inversion

The physical density is obtained by inverting the PyNeb emissivity ratio:

`R_{\mathrm{SII}} =`
`\frac{\epsilon_{6716}(T_e,n_e)}`
`{\epsilon_{6731}(T_e,n_e)} .`

The current product assumes:

```
Te = 10000 K
atom = PyNeb Atom("S", 2)
ratio = I(6716)/I(6731)
```

This gives the main HDU:

```
NE_SII
```

with units:

```
cm-3
```

3.4 Lookup table and exact high-density fallback

For speed, the script does not call `getTemDen` independently for every normal pixel. It creates or loads a reusable PyNeb lookup table in the current working directory:

```
pyneb_sii_6716_6731_density_lookup_te8000_13000_step1000_ne20_10000
0_n4096.npz
pyneb_sii_6716_6731_density_lookup_te8000_13000_step1000_ne20_10000
0_n4096.png
```

The cache stores:

```
temperature_grid = 8000, 9000, ..., 13000 K
density_grid     = geomspace(20, 100000, 4096) cm-3
ratio_grid_2d    = I(6716)/I(6731) from PyNeb emissivities
```

The density products currently use the 10000 K row of this table. The grid is logarithmic in density. With 4096 points from 20 to 100000 cm-3, the fractional density spacing is about:

`\exp\left[`
`\frac{\ln(100000/20)}{4095}`
`\right] - 1`
`\approx 0.0021 ,`

or about 0.2 percent per lookup step. This is much finer than the astrophysical uncertainty implied by the line-ratio errors.

The lookup is deliberately conservative:

- Ratios at or above the low-density limit are assigned the low-density floor.
- Ratios outside the high-density side of the cached range are not silently extrapolated.
- Any pixel whose interpolated density is non-finite or at least `1e4 cm-3` is recomputed with the exact PyNeb `getTemDen` call.

The exact-fallback threshold is therefore:

```
NE_SII >= 1e4 cm-3
```

The flag map is:

```
NE_SII_EXACT_HIDEN
```

where 1 means that the exact high-density PyNeb fallback was used.

3.5 Low-density floor at 20 cm-3

Brazzini et al. 2024 use [S II] 6716/6731 for density, and for regions below the low-density limit they impose:

`n_e = 20\ \mathrm{cm}^{-3}.`

The local implementation follows this logic. A pixel is flagged as fixed to 20 cm-3 if:

`R_{\mathrm{SII}} \ge R_{\mathrm{low-density}}` or if PyNeb returns a finite density below 20 cm-3. The final density is:

```
n_e =
\begin{cases}
20\ \mathrm{cm}^{-3}, & \mathrm{if\ below\ the\ low-density\ limit},\\
n_{e,\mathrm{PyNeb}}, & \mathrm{otherwise}.
\end{cases}
```

The flag map is:

```
NE_SII_FIXED20
```

where 1 means the final `NE_SII` value was fixed to 20 cm-3.

The uncertainty is:

```
NE_SII_ERR
```

computed by re-inverting the ratio at `R_SII - sigma(R_SII)` and `R_SII + sigma(R_SII)`, then taking half the difference:

```
\sigma(n_e) =
\frac{1}{2}
\left|
n_e(R+\sigma_R) - n_e(R-\sigma_R)
\right| .
```

For `NE_SII_FIXED20` pixels, the script sets:

```
\sigma(n_e) = 0 .
```

This encodes the adopted floor, not a measurement that the physical density is known exactly.

3.6 NE_SII_ALL

`NE_SII` is finite only when the [S II] doublet passes the detection gate. For some downstream operations it is useful to have a density everywhere in the spatially usable galaxy footprint. The script therefore also writes:

```
NE_SII_ALL
```

with:

`n_{e,\mathrm{all}} =`
`\begin{cases}`
`n_e([\mathrm{S\,II}]), & \mathrm{if}\ NE\_SII\ \mathrm{is\ finite},\\`
`20\ \mathrm{cm}^{-3}, & \mathrm{if\ the\ spatial\ pixel\ is\ usable},\\`
`\mathrm{NaN}, & \mathrm{if\ spatially\ masked}.`
`\end{cases}`

In code, "spatially usable" is `np.isfinite(V_STARS2)` after the spatial mask has been applied.

4. Common Te line ratios

All three metallicity methods use dust-corrected line intensities.

4.1 Te([N II])

The required [N II] temperature ratio is:

`R_{\mathrm{NII}} =`
`\frac{I(5755)}`
`{I(6548)+I(6583)} .`

The script evaluates:

```
PyNeb Atom("N", 2).getTemDen(
    R_NII,
    den=NE_SII,
    to_eval="L(5755)/(L(6548)+L(6584))",
)
```

and writes:

```
TE_NII_HII, TE_NII_HII_ERR
TE_NII_SF,  TE_NII_SF_ERR
```

Only temperatures in the range 1000 to 30000 K are retained.

The error is a finite-difference propagation in the line ratio and density:

```
\sigma(T_e)^2 =
\sigma(T_e; R)^2 + \sigma(T_e; n_e)^2 .
```

4.2 Te([S III])

The measured [S III] ratio is:

```
R_{\mathrm{SIII}} =
\frac{I(6312)}
{I(9069)+I(9532)} .
```

Because the current products use 9069 but not a robust measured 9532 map, the script adopts the theoretical relation:

```
I(9532) = 2.5\,I(9069).
```

Thus:

```
R_{\mathrm{SIII}} =
\frac{I(6312)}
{3.5\,I(9069)} .
```

The script evaluates:

```
PyNeb Atom("S", 3).getTemDen(
    R_SIII,
    den=NE_SII,
    to_eval="L(6312)/(L(9069)+2.5*L(9069))",
)
```

and writes:

```
TE_SIII_BR24_HII, TE_SIII_BR24_HII_ERR
TE_SIII_BR24_SF,  TE_SIII_BR24_SF_ERR
```

4.3 Inferred high-zone Te([O III])

MUSE does not normally cover the full [O III] auroral/nebular pair needed for a direct Te([O III]) measurement in the same way as classical optical direct methods. Brazzini et al. 2024 therefore infer the high-zone temperature from the [O III]-[S III] temperature relation, and state that the [O III]-[S III] relation is preferred over [O III]-[N II] because it is tighter.

The local code uses the implemented Brazzini-chain relations:

`T_e([\mathrm{S\,III}]) = 1.22\,T_e([\mathrm{N\,II}]) - 2000\ \mathrm{K},`

and:

`T_e([\mathrm{O\,III}]) = 0.80\,T_e([\mathrm{S\,III}]) + 2000\ \mathrm{K}.`

For the full Brazzini-style branch, `TE_OIII_BR24` is inferred from measured `TE_SIII_BR24`. For the NII+OII7325 branch, `TE_OIII_NII_CHAIN` is inferred from measured `TE_NII` through the two-step chain above.

The relation-error terms currently include the coefficient-error terms in the code plus intrinsic scatters of:

```
724 K   for Te([S III]) from Te([N II])
1270 K  for Te([O III]) from Te([S III])
```

5. Method 1: Brazzini+2024 multi-zone direct-style O/H

5.1 Output names

```
O_PLUS_BR24_HII,       O_PLUS_BR24_HII_ERR
O_DPLUS_BR24_HII,      O_DPLUS_BR24_HII_ERR
O_H_BR24_DIRECT_HII,   O_H_BR24_DIRECT_HII_ERR

O_PLUS_BR24_SF,        O_PLUS_BR24_SF_ERR
O_DPLUS_BR24_SF,       O_DPLUS_BR24_SF_ERR
O_H_BR24_DIRECT_SF,    O_H_BR24_DIRECT_SF_ERR

BR24_DIRECT_VALID_HII
BR24_DIRECT_VALID_SF
```

5.2 Required detections

For a pixel to enter this branch, all of the following must hold:

```
region mask is true: HII or SF
Hbeta detected
Halpha detected
SII6716 and SII6731 detected, giving finite NE_SII
NII5755 detected
NII6548+NII6583 detected
SIII6312 detected
SIII9069 detected
OII7318+7319+7329+7330 detected
OIII4959+5007 detected
```

Equivalently:

`M_{\mathrm{BR24}} =`
`M_{\mathrm{region}}`
`\land M_{\mathrm{Balmer}}`
`\land M_{n_e}`
`\land M_{T_{\mathrm{NII}}}`
`\land M_{T_{\mathrm{SIII}}}`
`\land M_{\mathrm{OII7325}}`
`\land M_{\mathrm{OIII}} .`

This is the strictest branch. It is normal for many galaxies to have zero valid spaxels because it requires both faint auroral systems and red [O II].

5.3 Temperatures used

The branch uses the multi-zone assignment described by Brazzini et al. 2024:

```
O+    uses measured Te([N II])
O++   uses Te([O III]) inferred from measured Te([S III])
```

The local implementation therefore sets:

`T_{\mathrm{low}} = T_e([\mathrm{N\,II}])`

and:

`T_{\mathrm{high}} =`
`T_e([\mathrm{O\,III}])`
`= 0.80\,T_e([\mathrm{S\,III}]) + 2000\ \mathrm{K}.`

5.4 Ionic abundances

The O+ intensity is:

`I([\mathrm{O\,II}]7325) = I(7318)+I(7319)+I(7329)+I(7330).`

The O++ intensity is:

`I([\mathrm{O\,III}]) = I(4959)+I(5007).`

PyNeb receives each line ratio with Hbeta normalized to 100:

`R_{\mathrm{OII}} = 100\frac{I([\mathrm{O\,II}]7325)}{I(\mathrm{H}\beta)}`

and:

`R_{\mathrm{OIII}} = 100\frac{I([\mathrm{O\,III}]4959+5007)}{I(\mathrm{H}\beta)} .`

Then:

```
O+/H+ = PyNeb Atom("O", 2).getIonAbundance(
    int_ratio=R_OII,
    tem=Te_NII,
    den=NE_SII,
    to_eval="L(7318)+L(7319)+L(7329)+L(7330)",
    Hbeta=100,
)
```

and:

```
O++/H+ = PyNeb Atom("O", 3).getIonAbundance(
    int_ratio=R_OIII,
    tem=Te_OIII_BR24,
    den=NE_SII,
    to_eval="L(4959)+L(5007)",
    Hbeta=100,
)
```

The total oxygen abundance is:

$$
\frac{O}{H} =
\frac{O^+}{H^+}
+
\frac{O^{++}}{H^+},
$$

and the map value is:

$$
12+\log(O/H) =
12 + \log_{10}
\left[
\frac{O^+}{H^+}
+
\frac{O^{++}}{H^+}
\right].
$$

This is the closest local analogue to the Brazzini et al. 2024 PHANGS-MUSE direct-method abundance workflow. The important difference is error handling: Brazzini et al. use Monte Carlo line-flux realizations, while the current local script uses deterministic finite-difference propagation.

6. Method 2: MAUVE NII+OII7325 semi-direct ionic-sum O/H

6.1 Output names

```
O_PLUS_NII_OII7325_HII,        O_PLUS_NII_OII7325_HII_ERR
O_DPLUS_NII_OII7325_HII,       O_DPLUS_NII_OII7325_HII_ERR
O_H_NII_OII7325_HII,           O_H_NII_OII7325_HII_ERR

O_PLUS_NII_OII7325_SF,         O_PLUS_NII_OII7325_SF_ERR
O_DPLUS_NII_OII7325_SF,        O_DPLUS_NII_OII7325_SF_ERR
O_H_NII_OII7325_SF,            O_H_NII_OII7325_SF_ERR

NII_OII7325_VALID_HII
NII_OII7325_VALID_SF
```

6.2 Required detections

This branch requires:

```
region mask is true: HII or SF
Hbeta detected
Halpha detected
SII6716 and SII6731 detected, giving finite NE_SII
NII5755 detected
NII6548+NII6583 detected
OII7318+7319+7329+7330 detected
OIII4959+5007 detected
```

It does not require [S III] 6312 or [S III] 9069. Therefore it should usually have more finite pixels than the full Brazzini-style branch.

6.3 Temperatures used

The low-zone temperature is measured:

```
T_{\mathrm{low}} =
T_e([\mathrm{N\,II}]).
```

The high-zone temperature is inferred through the local Brazzini-chain implementation:

```
T_e([\mathrm{S\,III}])_{\mathrm{chain}} =
1.22\,T_e([\mathrm{N\,II}]) - 2000\ \mathrm{K},
```

then:

```
T_e([\mathrm{O\,III}])_{\mathrm{chain}} =
0.80\,T_e([\mathrm{S\,III}])_{\mathrm{chain}}
+ 2000\ \mathrm{K}.
```

Combining those two equations:

```
T_e([\mathrm{O\,III}])_{\mathrm{chain}} =
0.976\,T_e([\mathrm{N\,II}]) + 400\ \mathrm{K}.
```

The code keeps the two-step form so that the uncertainty includes both relation error terms.

6.4 Ionic abundances and total abundance

The O+ abundance is computed from red [O II] 7325 with measured Te([N II]):

$$
\frac{O^+}{H^+}
=
f_{\mathrm{PyNeb}}
\left(
\frac{I(7318)+I(7319)+I(7329)+I(7330)}
{I(\mathrm{H}\beta)},
T_e([\mathrm{N\,II}]),
n_e
\right).
$$

The O++ abundance is computed from [O III] 4959+5007 with the inferred high-zone temperature:

$$
\frac{O^{++}}{H^+}
=
f_{\mathrm{PyNeb}}
\left(
\frac{I(4959)+I(5007)}
{I(\mathrm{H}\beta)},
T_e([\mathrm{O\,III}])_{\mathrm{chain}},
n_e
\right).
$$

The total abundance is the same ionic sum:

$$
12+\log(O/H) =
12 + \log_{10}
\left[
\frac{O^+}{H^+}
+
\frac{O^{++}}{H^+}
\right].
$$

This method is best described as a MAUVE-tailored semi-direct ionic-sum experiment. It is not a standard named PHANGS product. Its motivation is that Brazzini et al. 2024 already use Te([N II]) for O+ and red [O II] lines for the MUSE-accessible O+ abundance, while older work such as Pilyugin & Thuan 2007 also discusses using [O II] 7320,7330 when the blue [O II] 3727 doublet is not available. The main caveat is that the high-zone temperature is inferred from Te([N II]) rather than anchored by measured Te([S III]).

7. Method 3: Kreckel+2025 / Mendez-Delgado+2023 NII-only proxy

7.1 Output names

```
O_H_NII_K25_HII,  O_H_NII_K25_HII_ERR
O_H_NII_K25_SF,   O_H_NII_K25_SF_ERR

NII_K25_VALID_HII
NII_K25_VALID_SF
```

7.2 Required detections

This branch requires only the common Te inputs plus [N II]:

```
region mask is true: HII or SF
Hbeta detected
Halpha detected
SII6716 and SII6731 detected, giving finite NE_SII
NII5755 detected
NII6548+NII6583 detected
```

Then the script further requires the measured temperature to fall in the calibration range used by the implementation:

`8000\ \mathrm{K}`
`\le`
`T_e([\mathrm{N\,II}])`
`\le`
`13000\ \mathrm{K}.`

This extra temperature-range gate is why the integrated `O_H_NII_K25` value can be `nan` even when many spaxels have finite `TE_NII`: the integrated Te([N II]) summary must also be finite and within 8000-13000 K.

7.3 Metallicity equation

The implemented calibration is:

`12+\log(O/H) =`
`(-1.19)\,`
`\frac{T_e([\mathrm{N\,II}])}{10^4\ \mathrm{K}}`
`+ 9.68 .`

The uncertainty includes the Te uncertainty and the quoted slope/intercept calibration uncertainties:

```
\sigma_Z =
\sqrt{
\left(
\frac{1.19\,\sigma(T_e)}{10^4}
\right)^2
+
\left[
0.14\,
\frac{T_e([\mathrm{N\,II}])}{10^4}
\right]^2
+
0.15^2
}.
```

This branch does not compute O+ and O++ separately. It is a one-temperature oxygen-abundance proxy. Kreckel et al. 2025 motivate this because [N II] 5755 is the most commonly detected PHANGS-MUSE auroral line, and they show that Te([N II])-based gradients agree well with other metallicity-gradient measures. The calibration basis is tied to the Mendez-Delgado et al. 2023 temperature-metallicity relation, so its absolute scale should not be interpreted as the same thing as a classical collisionally excited-line ionic sum.

8. Error propagation summary

The local implementation is deterministic, not Monte Carlo. For a ratio:

```
R = \frac{A}{B},
```

it uses:

```
\sigma(R) =
|R|
\sqrt{
\left(\frac{\sigma_A}{A}\right)^2
+
\left(\frac{\sigma_B}{B}\right)^2
}.
```

For PyNeb quantities such as density, temperature, and ionic abundance, the script evaluates the quantity at the central value and at plus/minus one sigma in each driving variable, then uses:

$$
\sigma_Q =
\frac{1}{2}|Q_+ - Q_-|.
$$

Independent error contributions are added in quadrature.

For ionic-sum oxygen abundance:

$$
y =
\frac{O^+}{H^+}
+
\frac{O^{++}}{H^+},
$$

and:

$$
Z = 12+\log_{10}(y),
$$

the abundance uncertainty is:

$$
\sigma_Z =
\frac{
\sqrt{
\sigma(O^+/H^+)^2
+
\sigma(O^{++}/H^+)^2
}
}
{\ln(10)\,y}.
$$

This is fast and transparent, but it does not include all covariance terms and does not reproduce the full Monte Carlo procedure used by Brazzini et al. 2024. The local errors should therefore be treated as first-order propagated errors.

9. Integrated values printed to the log

For each of the HII and SF masks, the script prints integrated Te summaries:

```
NE_SII
Te(NII)
O_H_BR24_DIRECT
O_H_NII_OII7325
O_H_NII_K25
```

The integrated calculation is not the median of the spaxel maps. It does:

1. Selects the pixels that pass the relevant valid mask.
2. Sums raw line fluxes over those pixels.
3. Computes one integrated Balmer decrement from the summed Halpha and Hbeta.
4. Computes one integrated E(B-V).
5. Dust-corrects the integrated line fluxes.
6. Repeats the same PyNeb density, temperature, ionic-abundance, or K25 proxy equations on the integrated line ratios.

This is why a map can have many finite pixels while an integrated value is `nan`: the integrated branch still requires the summed-line ratios, integrated density, integrated temperature, and method-specific range gates to be valid. For `O_H_NII_K25`, the most common extra gate is:

`8000\ \mathrm{K}`

`\le`

`T_e([\mathrm{N\,II}])_{\mathrm{integrated}}`

`\le`

`13000\ \mathrm{K}.`

10. Practical interpretation of the valid-pixel counters

The log block:

```
PyNeb Te-method valid-pixel counts:
  NE_SII finite pixels: ...
  NE_SII fixed at 20 cm^-3: ...
  NE_SII exact high-density fallback pixels: ...
  NE_SII_ALL finite pixels: ...
  Te(NII) HII pixels: ...
  Te(NII) SF pixels: ...
  Brazzini+2024 direct HII pixels: ...
  Brazzini+2024 direct SF pixels: ...
  NII+OII7325 semi-direct HII pixels: ...
  NII+OII7325 semi-direct SF pixels: ...
  Kreckel+2025 NII-only HII pixels: ...
  Kreckel+2025 NII-only SF pixels: ...
```

should be read as follows:

- `NE_SII finite pixels`: pixels where the [S II] doublet passed detection and the final density is finite.
- `NE_SII fixed at 20 cm^-3`: subset of finite `NE_SII` pixels below the [S II] low-density limit or below the adopted 20 cm-3 floor.
- `NE_SII exact high-density fallback pixels`: pixels where interpolation was replaced by exact PyNeb `getTemDen`.
- `NE_SII_ALL finite pixels`: spatially usable pixels after the mask, using measured density when available and 20 cm-3 otherwise.
- `Te(NII)`: pixels where the common Te gate and [N II] temperature gate pass.
- `Brazzini+2024 direct`: pixels where [N II], [S III], [O II] 7325, and [O III] requirements all pass.
- `NII+OII7325 semi-direct`: pixels where [N II], [O II] 7325, and [O III] requirements pass, without requiring [S III].
- `Kreckel+2025 NII-only`: pixels where Te([N II]) is finite and in the 8000-13000 K calibration range.

11. References

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